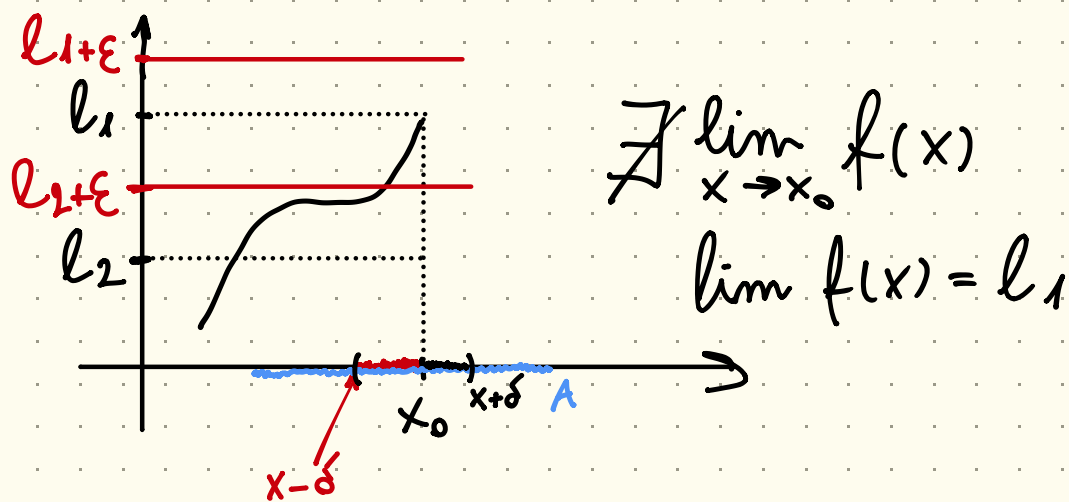
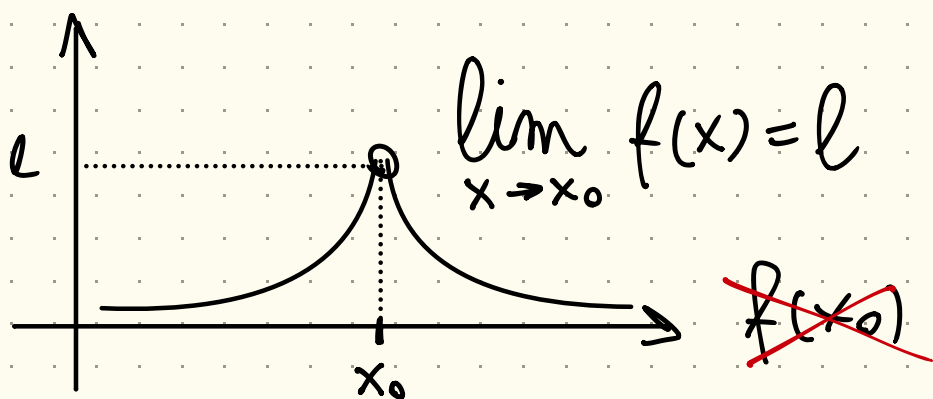


LIMITI SX E DX

• DEF.

$f: A \rightarrow \mathbb{R}$ SIA $x_0 \in \mathbb{R}$ T.C. $\forall \delta > 0$



$$(x_0 - \delta, x_0) \cap A \neq \emptyset \mid (x_0, x_0 + \delta) \cap A \neq \emptyset$$

SIA $l \in \mathbb{R}$. DICIAMO CHE

$$\lim_{x \rightarrow x_0^-} f(x) = l$$

$$\lim_{x \rightarrow x_0^+} f(x) = l$$

• SE $\forall \varepsilon > 0 \quad \exists \delta > 0$ T.C.

$$l - \varepsilon < f(x) < l + \varepsilon$$

• PER OGNI $x \in A$

$$x_0 - \delta < x < x_0 + \delta \quad | \quad x_0 < x < x_0 + \delta$$

• RICORDIAMO CHE:

$$\lim_{x \rightarrow x_0} f(x) = l : \quad x_0 - \delta < x < x_0 + \delta$$

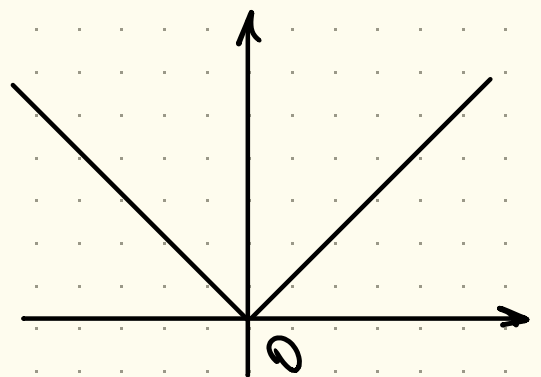
// INS FOTO.

es) MODULO

$$\lim_{x \rightarrow 0} |x| \stackrel{?}{=} |0| = 0$$

($\Rightarrow$ $|x|$ È CONTINUA IN 0)

$$|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$



QUANDO FACCO IL LIMITE DI UNA f CONT.
DEVO SEMPL. SOSTITUIRE IL VAL.

$$\bullet \lim_{x \rightarrow 0^-} |x| = \lim_{x \rightarrow 0^-} (-x) = -0 = 0$$

$\uparrow$
 $-x$ CONT.

$$\bullet \lim_{x \rightarrow 0^+} |x| = \lim_{x \rightarrow 0^+} x = 0 = 0$$

$\uparrow$
 x CONT.

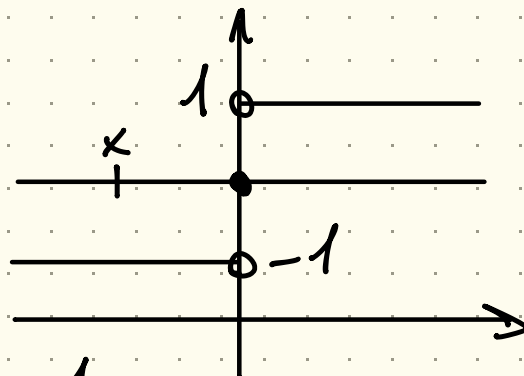
TEO $\Rightarrow \lim_{x \rightarrow 0} |x| = 0$

DS) SEGNO

$$\bullet \lim_{x \rightarrow 0} \text{SGN}(x)$$

$$\bullet \lim_{x \rightarrow 0^-} \text{SGN}(x) = \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (-1) = -1$$

$$\bullet \lim_{x \rightarrow 0^+} \text{SGN}(x) = \lim_{x \rightarrow 0^+} 1 = 1$$



$\Rightarrow \nexists \lim_{x \rightarrow 0} \text{SGN}(x)$

$$\bullet f(x) = \begin{cases} e^{x-1} - x^2 - 1 & x < 1 \\ \cos(\pi x) & x \geq 1 \end{cases}$$

• QUAL È L'INSIEME DI CONTINUITÀ DI f ?

- f È CONTINUA IN $(-\infty, 1)$ E IN $(1, -\infty)$

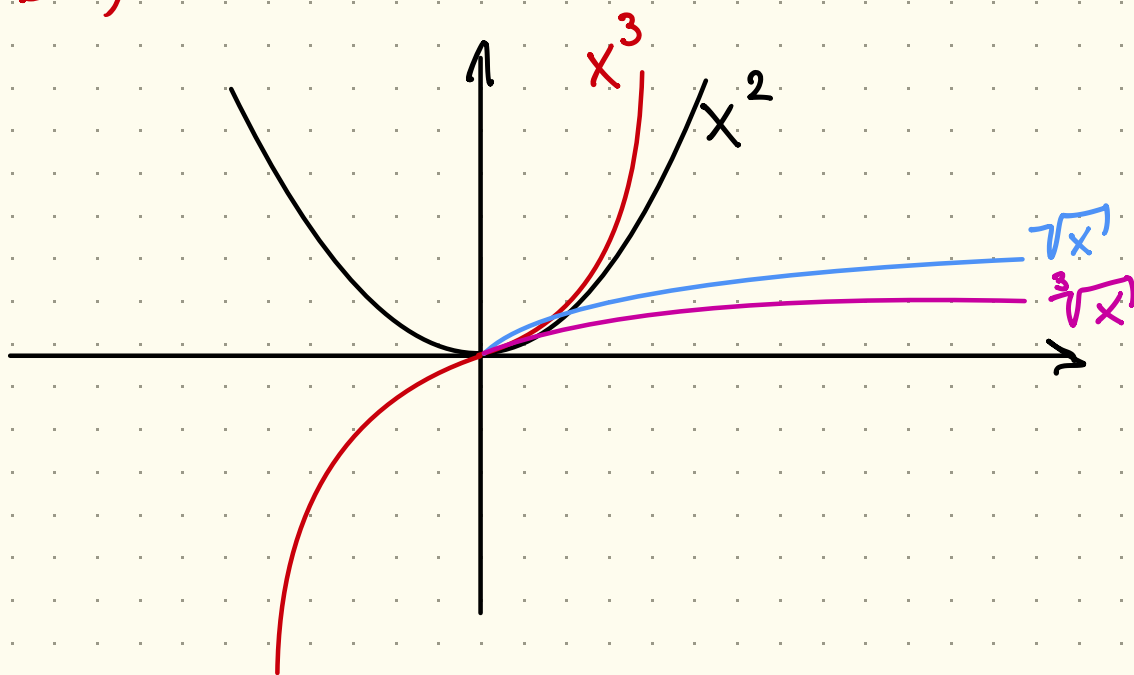
PERCHÉ SOMMA E COMP. DI f CONT.

STUDIAMO LA CONTINUITÀ IN $\leq \frac{1}{-1}$

f È CONTINUA IN $1 \Leftrightarrow$

// APPUNTI MANCANTI

es)



$$\left. \begin{array}{l} \lim_{x \rightarrow +\infty} x^2 = +\infty \\ \lim_{x \rightarrow -\infty} x^2 = +\infty \end{array} \right\} x^2$$

$$\bullet \lim_{x \rightarrow \pm \infty} x^3 = \pm \infty \quad x^3$$

$$\bullet \lim_{x \rightarrow +\infty} \sqrt{x} = +\infty \quad \sqrt{x} \quad \nexists \lim_{x \rightarrow -\infty} \sqrt{x}$$

$$\bullet \lim_{x \rightarrow \pm \infty} \sqrt[3]{x} = \pm \infty \quad \sqrt[3]{x}$$

→ POICHÉ $-\infty$ NON É UN PTO DI ACCUMULAZIONE PER $\text{DOM } \sqrt{x} = [0, +\infty)$

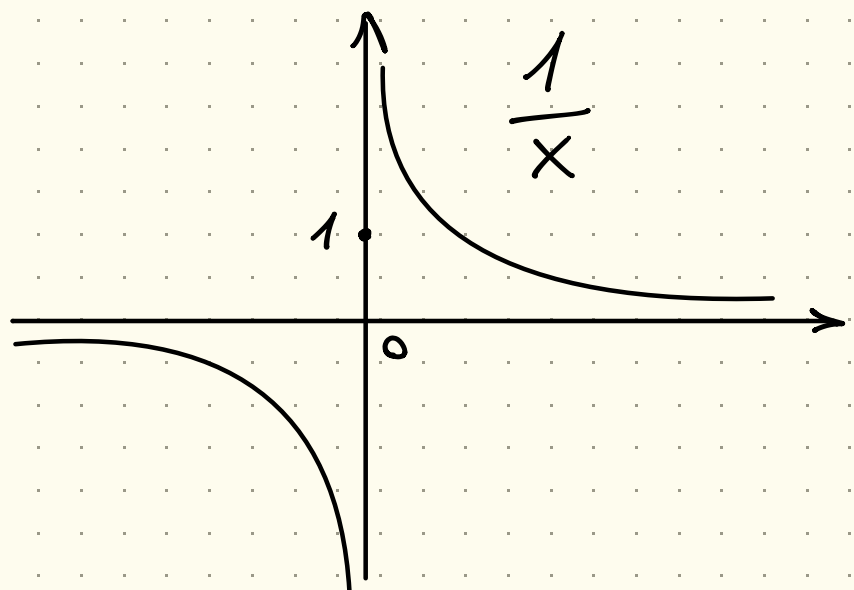
$$\bullet \lim_{x \rightarrow 0} \frac{1}{x^2} = +\infty$$

$$\bullet \lim_{x \rightarrow \pm \infty} \frac{1}{x^2} = 0$$

$$\bullet \lim_{x \rightarrow \pm \infty} \frac{1}{x} = 0$$

$$\bullet \nexists \lim_{x \rightarrow 0} \frac{1}{x}$$

$$\bullet \lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty \quad x^a \quad a \in \mathbb{R}$$



- $\lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty$

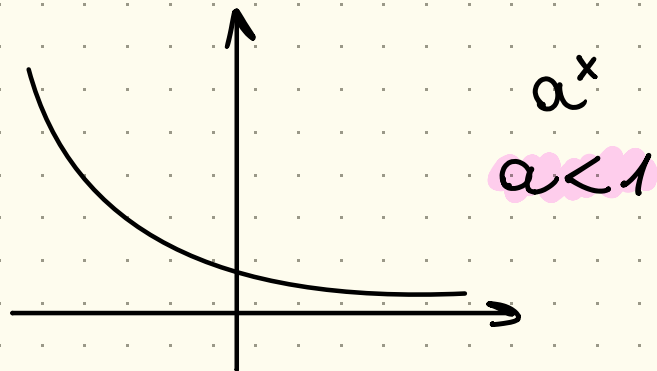
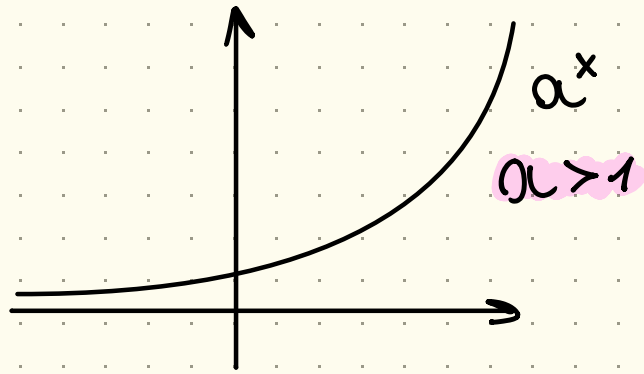
//ESP.

- $\lim_{x \rightarrow -\infty} a^x = 0$

- $\lim_{x \rightarrow +\infty} a^x = +\infty$

- $\lim_{x \rightarrow -\infty} a^x = +\infty$

- $\lim_{x \rightarrow +\infty} a^x = 0$

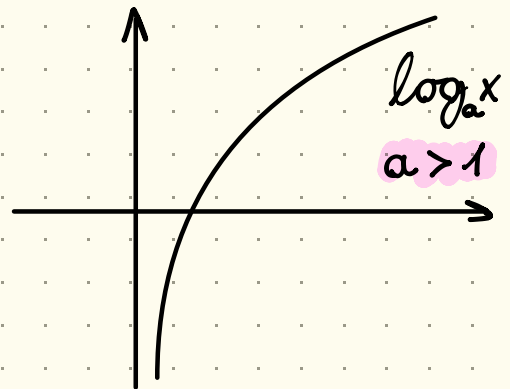


• LOGARITMO

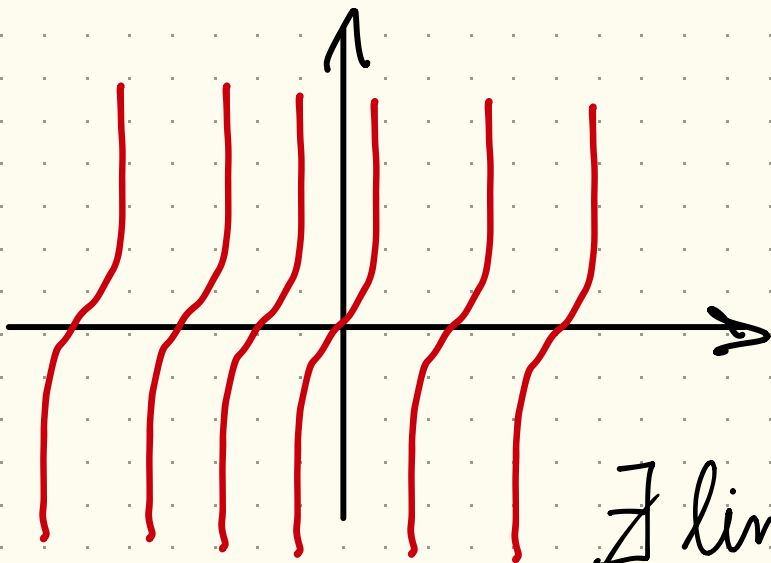
Dom $\log_a = (0, +\infty)$

- $\lim_{x \rightarrow 0} \log_a x = -\infty$

- $\lim_{x \rightarrow +\infty} \log_a x = +\infty$



• TANGENTE



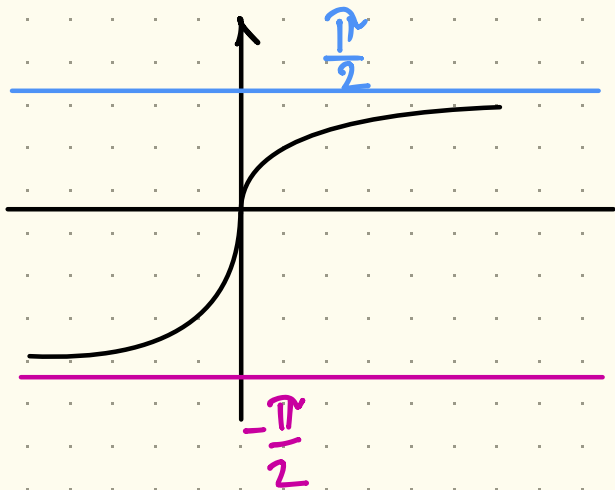
$$\nexists \lim_{x \rightarrow +\infty} \tan x$$

$$\lim_{x \rightarrow -\frac{\pi}{2}^+} \tan x = -\infty$$

$$\lim_{x \rightarrow -\frac{\pi}{2}^-} \tan x = +\infty$$

$$\lim_{x \rightarrow \frac{\pi}{2}^-} \tan x = +\infty$$

$$\lim_{x \rightarrow \frac{\pi}{2}^+} \tan x = -\infty$$



$$\lim_{x \rightarrow +\infty} \arctan x = \frac{\pi}{2}$$

$$\lim_{x \rightarrow -\infty} \arctan x = -\frac{\pi}{2}$$

$$\lim_{x \rightarrow +\infty} x^2 + e^x$$

$$\lim_{x \rightarrow +\infty} (x^2 - 1) \cdot (a \tan x)$$

Teo (algebra dei limiti)

$f, g: A \rightarrow \mathbb{R}$, x_0 di acc. per A .

Supp che

$$\exists \lim_{x \rightarrow x_0} f(x) = l_1 \in [-\infty, +\infty]$$

$$\exists \lim_{x \rightarrow x_0} g(x) = l_2 \in [-\infty, +\infty]$$

Allora:

a) LIMITE DELLA SOMMA

$$\lim_{x \rightarrow x_0} f(x) + g(x) = l_1 + l_2$$

$$\lim_{x \rightarrow x_0} f(x) + g(x) =$$

	$l_1 \in \mathbb{R}$	$l_1 = -\infty$	$l_1 = +\infty$
$l_2 \in \mathbb{R}$	$l_1 + l_2$	$-\infty$	$+\infty$
$l_2 = -\infty$	$-\infty$	$-\infty$	? = f.i.
$l_2 = +\infty$	$+\infty$	? = f.i.	$+\infty$

$$\text{es } \lim_{x \rightarrow +\infty} \underbrace{a \tan x}_{\rightarrow \frac{\pi}{2}} + \underbrace{\left(\frac{1}{2}\right)^x}_{\rightarrow 0} = \frac{\pi}{2} + 0$$

$$\lim_{x \rightarrow +\infty} x^2 + e^x = +\infty$$

$$\lim_{x \rightarrow -\infty} a \tan x + x^2 = +\infty$$

$$\lim_{x \rightarrow -\infty} \underbrace{x^2 + x^3}_{\rightarrow +\infty \rightarrow -\infty} = ?$$

b) LIMITE DEL PRODOTTO

$$\lim_{x \rightarrow x_0} f(x) g(x) =$$

	$l_1 < 0$	$l_1 = 0$	$l_1 > 0$	$l_1 = +\infty$	$l_1 = -\infty$
$l_2 < 0$				$-\infty$	$+\infty$
$l_2 = 0$	$l_1 \cdot l_2$			?	?
$l_2 > 0$				$+\infty$	$-\infty$
$l_2 = +\infty$	$-\infty$	?	$+\infty$	$+\infty$	$-\infty$
$l_2 = -\infty$	$+\infty$	?	$-\infty$	$-\infty$	$+\infty$